Chengdu Ebyte Electronic Technology Co.,Ltd

Wireless Modem

User Manual

M31-U-PN Series Profinet Distributed IO Host

Contents

1 Product Overview

1.1 Product Introduction

The M31-U-PN series distributed I/O master is a slave module supporting the Profinet protocol . It maps I/O data to Profinet. Integration is achieved using GSD files, allowing PLCs to read and write I/O ports without requiring code. It supports various Siemens PLC models, including S7-200smart, S7-300, S7-1200, S7-1500, and S7-1500. It features two network ports, supports switch functionality, and integrates a Profinet-to-Modbus RTU gateway.

The device features a modular design, allowing users to flexibly expand it according to their actual needs. It supports multiple I/O types , and when the existing configuration is insufficient to meet specific application requirements, users can easily add matching I/O expansion modules without replacing the entire system, effectively saving costs and simplifying the on-site deployment process.

This product allows connection to up to 16 IO expansion modules (including the host itself) and is built in accordance with EMC Level 3 protection standards. IO input/output and power supply are fully isolated, providing excellent electromagnetic compatibility, outstanding performance and high reliability, and meets CE certification standards.

1.2 Features

- It uses the standard Profinet protocol for communication and can be networked with PLCs, configuration files, and host computers.
- Dual industrial Ethernet ports 10/100M auto-sensing, with built-in switch functionality
- Supports Profinet to Modbus RTU gateway function
- Modular splicing structure, small size, compact structure
- Supports differential analog input, 16-bit resolution, and accuracy within 1‰.
- Millisecond response time, with internal bus speeds up to 1ms.
- It can connect up to 16 I/O expansion modules (including the host itself) .
- The maximum number of I/O operations can be supported up to 256 bits.
- Arbitrary combinations of digital input/output, analog input/output, etc.
- Supports multiple I/O types including DI (NPN/PNP), AI (single-ended current), AI (differential current), AI (differential voltage), DO (PNP), DO (relay), AO (current), AO (voltage) , and PT100 RTD .
- It adopts industrial-grade hardware design and has multiple protections including electrostatic discharge, lightning surge, fast pulse group , and reverse power connection protection.

- Features I/O input/output, RS485 isolation, and power isolation.
- Supports mounting via positioning holes and guide rails.
- Meets CE certification standards

1.3 Product Model List

Product Model	Product Specifications	digital input DI	Analog input AI	Switch output DO	Analog output AO
M31-AFAX4440G-U-PN	4DI+4DO+4AI	4 (NPN, PNP)	4 (Differential Current)	4 (Relay)	—
M31-AXAX8080G-U-PN	8DI+8DO	8 (NPN, PNP)	—	8 (Relay)	—
M31-AXXX8000G-U-PN	8DI	8 (NPN, PNP)	—	—	—
M31-AXXXA000G-U-PN	16 DI	16 (NPN, PNP)	—	—	—
M31-AXEX8080G-U-PN	8DI+8DO	8 (NPN, PNP)	—	8 (Transistors)	—
M31-XXAX00A0G-U-PN	16 DO	—	—	16 (Relay)	—
M31-XXEX00A0G-U-PN	16 DO	—	—	16 (Transistors)	—
M31-AXAX4040G-U-PN	4 DI+ 4 DO	4 (NPN, PNP)	—	4 (Relay)	—
M31-XFXX0800G-U-PN	8AI	—	8 (Differential Current)	—	—
M31-XAXX0A00G-U-PN	16AI	—	16 (Single-ended current)	—	—
M31-XBXX0A00G-U-PN	16AI	—	16 (Single-ended voltage)	—	—
M31-XXXA0008G-U-PN	8AO	—	—	—	8 (Current)
M31-XXXB0008G-U-PN	8AO	—	—	—	8 (Voltage)
M31-XDXX0600G-U-PN	6PT100	—	PT100 RTD	—	—

Please consult the official website for details on expansion module models.

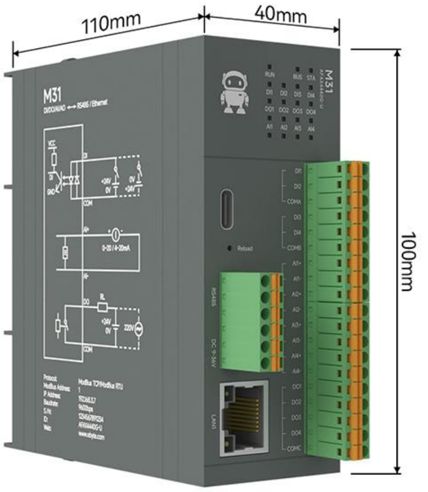
2 Technical indicators

2.1 Specifications

Profinet Distributed I/O Host		
category	**name**	**parameter**
power supply	Operating voltage	DC 9 ～ 36V ; with reverse polarity protection
	Power indicator	RUN red LED indicator
	Power consumption	2-4W
communication	Communication interface	RJ45 *2 (dual network ports) , RS485
	Communication Protocol	Standard Profinet (RT) protocol
	Internal bus	Up to 1ms
DI Input	Input type	NPN, PNP
	Input range	DC 10 ~ 28V
	Isolation methods	Each channel is individually opto-isolated.
	Input impedance	7.2kΩ
	sampling frequency	1000 Hz
	Counting frequency	<500Hz
	Filtering time	Default time 6ms
	Input Instructions	DI green LED indicator
AI input	Acquisition characteristics	Single-ended input/differential input (model optional)
	Input type	Single-ended current: 0-20mA, 4-20mA ; Differential current: 0-20mA, 4-20mA, ±20mA; Differential voltage: 0-5V, ±5V, 0-10V, ±10V; Single-ended voltage: 0-5V, 0-10V
	AI resolution	16-bit (differential)
	AI accuracy	1‰ (differential) / 3‰ (single-ended)
	refresh rate	Maximum 70Hz
	Input Instructions	AI Green LED Indicator
DO (relay) output	DO output type	Type A relay (normally open)
	DO output mode	Level output , pulse output
	Relay contact capacity	5A 30VDC, 5A 250VAC (maximum total current supported at the same COM common

		terminal is 8A)
	Output Indicator	DO green LED indicator
DO (PNP) output	DO output type	PNP type
	DO output mode	Level output , pulse output
	Output capability	Supports output up to 0.5A per channel and 4A per COM port at voltages from 10 to 30V.
	Output Indicator	DO green LED indicator
AO output	Output type	0-20mA, 4-20mA / 0-5V , 0-10V (Voltage and current models are optional)
	Output accuracy	1‰
	Output impedance	AOV: 40R AOI: 656kΩ
	Output Indicator	AO green LED indicator
PT100 input	Input type	2/3 wire PT100 RTD
	Temperature measurement range	-200～850℃
	Temperature resolution	0.1℃
	accuracy	±0.1%±1℃
	refresh rate	Maximum 70Hz
	Input Instructions	RTD Green LED Indicator
other	Product Dimensions	110mm * 40mm * 100mm (Length * Width * Height)
	Operating temperature and humidity	-40 to +85℃, 5% to 95%RH (non-condensing)
	Storage temperature and humidity	-40 to +105℃, 5% to 95%RH (non-condensing)
	Installation method	Positioning holes and guide rail installation

2.2 Dimensions



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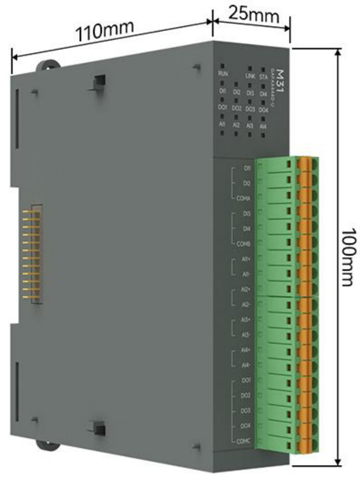


扩展模块

2.3 Port, button and LED indicator description

2.3.1 M31-AFAX4440G-U-PN

M31-AFAX4440G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DI1	DI1 digital input	DI1 digital input interface, used in conjunction with COM A
DI2	DI2 digital input	DI2 digital input interface, used in conjunction with COM A
COM A	DI digital input common terminal	DI1-DI2 share COM A common terminal
DI3	DI3 digital input	DI3 digital input interface, used in conjunction with COM B
DI4	DI4 digital input	DI4 digital input interface, used in conjunction with COM B
COM B	DI digital input common terminal	DI3-DI4 share a common COM B terminal
DO1	DO1 digital output	DO1 is a digital output interface used in conjunction with COMC.
DO2	DO2 digital output	DO2 digital output interface, used in conjunction with COMC
DO3	DO3 digital output	DO3 digital output interface, used in conjunction with COMC
DO4	DO4 digital output	DO4 digital output interface, used in conjunction with COMC
COM C	DO's COM side	DO1-DO4 share a common COMC for use.
AI1+	AI1 Analog Input +	AI1 analog input + interface, used in conjunction with AI1-
AI1-	AI1 Analog Input -	AI1 Analog Input Interface, for use with AI1+

AI2+	AI2 analog input +	AI2 analog input + interface, used in conjunction with AI2-
AI2-	AI2 Analog Input -	AI2 Analog Input Interface, for use with AI2+
AI3+	AI3 Analog Input +	AI3 analog input + interface, used in conjunction with AI3-
AI3-	AI3 Analog Input -	AI3 Analog Input Interface, for use with AI3+
AI4+	AI4 Analog Input +	AI4 analog input + interface, used in conjunction with AI4-
AI4-	AI4 Analog Input -	AI4 Analog Input Interface, for use with AI4+
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-AFAX4440G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Equipment operating status indicator lights	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
DI1	DI1 Input Indicator	Green LED light; On: DI1 valid input; Off: DI1 invalid input.
DI2	DI2 Input Indicator	Green LED light; On: DI2 input is valid; Off: DI2 input is invalid.
DI3	DI3 Input Indicator	Green LED light; On: DI3 input is valid; Off: DI3 input is invalid.
DI4	DI4 Input Indicator	Green LED light; On: DI4 input is valid; Off: DI4 input is invalid.
DO1	DO1 output indicator light	Green LED light; On: DO1 relay closed; Off: DO1 relay open.
DO2	DO2 output indicator light	Green LED light; On: DO2 relay closed; Off: DO2 relay open.
DO3	DO3 output indicator light	Green LED light; On: DO3 relay closed; Off: DO3 relay open.

DO4	DO4 output indicator light	Green LED light; On: DO4 relay closed; Off: DO4 relay open.
AI1	AI1 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI2	AI2 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI3	AI3 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI4	AI4 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.

2.3.2 M31-AXAX8080G-U-PN

M31-AXAX8080G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DO1	DI1 digital input	DI1 digital input interface, used in conjunction with COMA
DI2	DI2 digital input	DI2 digital input interface, used in conjunction with COMA
DI3	DI3 digital input	DI3 digital input interface, used in conjunction with COMA
DI4	DI4 digital input	DI4 digital input interface, used in conjunction with COMA
COMA	DI digital input common terminal	DI1-DI4 share the same COMA common terminal
DI5	DI5 digital input	DI5 digital input interface, used in conjunction with COMB.
DI6	DI6 digital input	The DI6 digital input interface is used in conjunction with the COMB module.
DI7	DI7 digital input	The DI7 digital input interface is used in conjunction with the COMB module.
DI8	DI8 digital input	The DI8 digital input interface is used in conjunction with the COMB module.
COMB	DI digital input common terminal	DI5-DI8 share a common COMP terminal
DO1	DO1 digital output	DO1 is a digital output interface used in conjunction with COMC.
DO2	DO2 digital output	DO2 digital output interface, used in conjunction with COMC
DO3	DO3 digital output	DO3 digital output interface, used in conjunction with COMC
DO4	DO4 digital output	DO4 is a digital output interface used in conjunction with COMC.
DO5	DO5 digital output	DO5 digital output interface, used in conjunction with COMC
DO6	DO6 digital output	DO6 digital output interface, used in conjunction with COMC
DO7	DO7 digital output	DO7 digital output interface, used in conjunction with COMC
DO8	DO8 digital output	DO8 digital output interface, used in conjunction with COMC
COMC	DO's COM side	DO1-DO8 share a common COMC for coordinated use

Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-AXAX8080G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Equipment operating status indicator lights	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flashing: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
DI1	DI1 Input Indicator	Green LED light; On: DI1 valid input; Off: DI1 invalid input.
DI2	DI2 Input Indicator	Green LED light; On: DI2 input is valid; Off: DI2 input is invalid.
DI3	DI3 Input Indicator	Green LED light; On: DI3 input is valid; Off: DI3 input is invalid.
DI4	DI4 Input Indicator	Green LED light; On: DI4 input is valid; Off: DI4 input is invalid.
DI5	DI5 Input Indicator	Green LED light; On: DI5 input is valid; Off: DI5 input is invalid.
DI6	DI6 Input Indicator	Green LED light; On: DI6 input is valid; Off: DI6 input is invalid.
DI7	DI7 Input Indicator	Green LED light; On: DI7 valid input; Off: DI7 invalid input.
DI8	DI8 Input Indicator	Green LED light; On: DI8 input is valid; Off: DI8 input is invalid.
DO1	DO1 output indicator light	Green LED light; On: DO1 relay closed; Off: DO1 relay open.
DO2	DO2 output indicator light	Green LED light; On: DO2 relay closed; Off: DO2 relay open.
DO3	DO3 output indicator light	Green LED light; On: DO3 relay closed; Off: DO3 relay open.
DO4	DO4 output indicator	Green LED light; On: DO4 relay closed; Off: DO4 relay open.

	light	
DO5	DO5 output indicator light	Green LED light; On: DO5 relay closed; Off: DO5 relay open.
DO6	DO6 output indicator light	Green LED light; On: DO6 relay closed; Off: DO6 relay open.
DO7	DO7 output indicator light	Green LED light; On: DO7 relay closed; Off: DO7 relay open.
DO8	DO8 output indicator light	Green LED light; On: DO8 relay closed; Off: DO8 relay open.

2.3.3 M31-AXXX8000G-U-PN

M31-AXXX8000G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DI1	DI1 digital input	DI1 is a digital input interface used in conjunction with COM 1.
COM1	COM port of D I1	DI1 common terminal of digital input interface
DI 2	DI 2 digital input	DI 2 is a digital input interface used in conjunction with COM 2.
COM2	DI2 's COM port	The common terminal of the DI2 digital input interface
DI 3	DI 3 digital input	DI 3 is a digital input interface used in conjunction with COM 3 .
COM3	DI3 's COM port	The common terminal of the DI3 digital input interface
DI 4	DI 4 digital input	DI 4 is a digital input interface used in conjunction with COM 4 .
COM4	DI4 's COM port	The common terminal of the DI4 digital input interface
DI 5	DI 5 digital input	DI 5 is a digital input interface used in conjunction with COM 5 .
COM5	DI5 COM terminal	The common terminal of the DI5 digital input interface
DI 6	DI 6 digital input	DI 6 digital input interface, used in conjunction with COM 6
COM6	DI61 COM port	The common terminal of the DI6 digital input interface
DI 7	DI 7 digital input	DI 7 is a digital input interface used in conjunction with COM 7 .
COM7	DI7 's COM port	The common terminal of the DI7 digital input interface
DI 8	DI 8 digital input	DI 8 digital input interface, used in conjunction with COM 8
COM8	DI8 's COM port	The common terminal of the DI8 digital input interface
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface

G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-AXXX8000G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
DI 1	DI1 Input Indicator	Green LED light; On: DI1 valid input; Off: DI1 invalid input.
DI 2	DI 2 Input Indicator	Green LED light; On: DI 2 valid input; Off: DI 2 invalid input.
DI 3	DI 3 Input Indicator	Green LED light; On: DI 3 valid input; Off: DI 3 invalid input.
DI 4	DI 4 Input Indicator	Green LED light; On: DI 4 valid input; Off: DI 4 invalid input.
DI 5	DI 5 Input Indicator	Green LED light; On: DI 5 valid input; Off: DI 5 invalid input.
DI 6	DI 6 Input Indicator	Green LED light; On: DI 6 valid input; Off: DI 6 invalid input.
DI 7	DI 7 Input Indicator	Green LED light; On: DI 7 valid input; Off: DI 7 invalid input.
DI 8	DI 8 Input Indicator	Green LED light; On: DI 8 valid input; Off: DI 8 invalid input.

2.3.4 M31-AXXXA000G-U-PN

M31-AXXXA000G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DI 1	DI 1 digital input	DI1 digital input interface, used in conjunction with COM A
DI 2	DI 2 digital input	DI 2 is a digital input interface used in conjunction with COM A.
DI 3	DI 3 digital input	DI 3 digital input interface, used in conjunction with COM A
DI 4	DI 4 digital input	DI 4 digital input interface, used in conjunction with COM A
DI 5	DI 5 digital input	DI 5 digital input interface, used in conjunction with COM A

DI 6	DI 6 digital input	DI 6 digital input interface, used in conjunction with COM A
DI 7	DI 7 digital input	DI 7 digital input interface, used in conjunction with COM A
DI 8	DI 8 digital input	DI 8 digital input interface, used in conjunction with COM A
COMA	DI digital input common terminal	DI1-DI8 share the same COMA common terminal
DI 9	DI 9 digital input	DI 9 digital input interface, used in conjunction with COM A
DI1 0	DI1 0 digital input	DI 10 digital input interface, used in conjunction with COM A
DI1 1	DI1 1- digit input	DI1 is a digital input interface used in conjunction with COM A.
DI1 2	DI1 2 digital input	DI1 2 is a digital input interface used in conjunction with COM A.
DI1 3	DI1 3 digital input	DI1 3 digital input interface, used in conjunction with COM A
DI1 4	DI1 4- digit digital input	DI1 4 digital input interface, used in conjunction with COM A
DI1 5	DI1 5 digital input	DI1 5 digital input interface, used in conjunction with COM A
DI1 6	DI1 6- digit digital input	DI1 6 digital input interface, used in conjunction with COM A
COMB	DI digital input common terminal	DI 9 - DI 16 share a common COM B terminal
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-AXXXA000G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication	Blue LED light; flashing: indicates data exchange between the host and the

	indicator	expansion module; off: indicates no data exchange between the host and the expansion module.
DI1	DI1 Input Indicator	Green LED light; On: DI1 valid input; Off: DI1 invalid input.
DI2	DI2 Input Indicator	Green LED light; On: DI2 input is valid; Off: DI2 input is invalid.
DI3	DI3 Input Indicator	Green LED light; On: DI3 input is valid; Off: DI3 input is invalid.
DI4	DI4 Input Indicator	Green LED light; On: DI4 input is valid; Off: DI4 input is invalid.
DI5	DI5 Input Indicator	Green LED light; On: DI5 input is valid; Off: DI5 input is invalid.
DI6	DI6 Input Indicator	Green LED light; On: DI6 input is valid; Off: DI6 input is invalid.
DI7	DI7 Input Indicator	Green LED light; On: DI7 valid input; Off: DI7 invalid input.
DI8	DI8 Input Indicator	Green LED light; On: DI8 input is valid; Off: DI8 input is invalid.
DI 9	DI 9 Input Indicator	Green LED light; On: DI 9 valid input; Off: DI 9 invalid input.
DI 10	DI 10 Input Indicator	Green LED light; On: DI 10 valid input; Off: DI 10 invalid input.
DI 11	DI 11 Input Indicator	Green LED light; On: DI 11 valid input; Off: DI 11 invalid input.
DI 12	DI 12 Input Indicator	Green LED light; On: DI 12 valid input; Off: DI 12 invalid input.
DI 13	DI 13 Input Indicator	Green LED light; On: DI 13 valid input; Off: DI 13 invalid input.
DI 14	DI 14 Input Indicator	Green LED light; On: DI 14 valid input; Off: DI 14 invalid input.
DI 15	DI 15 Input Indicator	Green LED light; On: DI 15 valid input; Off: DI 15 invalid input.
DI 16	DI 16 Input Indicator	Green LED light; On: D16 valid input; Off: D16 invalid input.

2.3.5 M31-AXEX8080G-U-PN

M31-AXEX8080G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DI 1	DI1 digital input	DI1 digital input interface, used in conjunction with COMA
DI2	DI2 digital input	DI2 digital input interface, used in conjunction with COMA
DI3	DI3 digital input	DI3 digital input interface, used in conjunction with COMA
DI4	DI4 digital input	DI4 digital input interface, used in conjunction with COMA
COMA	DI digital input common terminal	DI1-DI4 share the same COMA common terminal
DI5	DI5 digital input	DI5 digital input interface, used in conjunction with COMB.
DI6	DI6 digital input	The DI6 digital input interface is used in conjunction with the COMB module.
DI7	DI7 digital input	DI7 digital input interface, used in conjunction with COMB.
DI8	DI8 digital input	The DI8 digital input interface is used in conjunction with the COMB module.
COMB	DI digital input common terminal	DI5-DI8 share a common COMB terminal
DO1	DO1 digital output	DO1 is a digital output interface used in conjunction with COMC.
DO2	DO2 digital output	DO2 digital output interface, used in conjunction with COMC

DO3	DO3 digital output	DO3 digital output interface, used in conjunction with COMC
DO4	DO4 digital output	DO4 digital output interface, used in conjunction with COMC
DO5	DO5 digital output	DO5 digital output interface, used in conjunction with COMC
DO6	DO6 digital output	DO6 digital output interface, used in conjunction with COMC
DO7	DO7 digital output	DO7 digital output interface, used in conjunction with COMC
DO8	DO8 digital output	DO8 digital output interface, used in conjunction with COMC
COMC	DO's COM port	DO1-DO8 share a common COMC for coordinated use
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-AXEX8080G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
DI1	DI1 Input Indicator	Green LED light; On: DI1 valid input; Off: DI1 invalid input.
DI2	DI2 Input Indicator	Green LED light; On: DI2 input is valid; Off: DI2 input is invalid.
DI3	DI3 Input Indicator	Green LED light; On: DI3 input is valid; Off: DI3 input is invalid.
DI4	DI4 Input Indicator	Green LED light; On: DI4 input is valid; Off: DI4 input is invalid.
DI5	DI5 Input Indicator	Green LED light; On: DI5 input is valid; Off: DI5 input is invalid.
DI6	DI6 Input Indicator	Green LED light; On: DI6 input is valid; Off: DI6 input is invalid.
DI7	DI7 Input Indicator	Green LED light; On: DI7 valid input; Off: DI7 invalid input.
DI8	DI8 Input Indicator	Green LED light; On: DI8 input is valid; Off: DI8 input is invalid.
DO1	DO1 output indicator	Green LED light; On: DO1 transistor is conducting; Off: DO1 transistor is

	light	cut off.
DO2	DO2 output indicator light	Green LED light; On: DO 2 transistor is conducting; Off: DO 2 transistor is cut off.
DO3	DO3 output indicator light	Green LED light; On: DO 3 transistor is conducting; Off: DO 3 transistor is cut off.
DO4	DO4 output indicator light	Green LED light; On: DO 4 transistor is conducting; Off: DO 4 transistor is cut off.
DO5	DO5 output indicator light	Green LED light; On: DO 5 transistor is conducting; Off: DO 5 transistor is cut off.
DO6	DO6 output indicator light	Green LED light; On: DO 6 transistor is conducting; Off: DO 6 transistor is cut off.
DO7	DO7 output indicator light	Green LED light; On: DO 7 transistor is conducting; Off: DO 7 transistor is cut off.
DO8	DO8 output indicator light	Green LED light; On: DO 8 transistor is conducting; Off: DO 8 transistor is cut off.

2.3.6 M31-XXAX00A0G-U-PN

M31-XXAX00A0G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DO1	DO1 digital output	DO1 digital output interface, used in conjunction with COMA
DO2	DO 2 digital output	DO 2 is a digital output interface used in conjunction with COMA.
DO3	DO 3 digital output	DO 3 digital output interface, used in conjunction with COMA
DO4	DO 4 digital output	DO 4 digital output interface, used in conjunction with COMA
DO5	DO 5 digital output	DO 5 digital output interface, used in conjunction with COMA
DO 6	DO 6 digital output	DO 6 digital output interface, used in conjunction with COMA
DO7	DO 7 digital output	DO 7 digital output interface, used in conjunction with COMA
DO8	DO 8 digital output	DO 8 digital output interface, used in conjunction with COMA
COM A	DO's COM side	DO1 - DO8 share COM A common terminal
DO 9	DO 9 digital output	DO 9 digital output interface, used in conjunction with COM 8
DO 10	DO 10 digital output	DO 10 digital output interface, used in conjunction with COM B
DO 11	DO1 1- way digital output	DO1 is a digital output interface for use with COM B.
DO 12	DO1 2 digital output	DO1 2 -bit digital output interface, used in conjunction with COM B
DO 13	DO1 3- way digital output	DO1 3 digital output interface, used in conjunction with COM B
DO 14	DO1 4- way digital output	DO1 4 -way digital output interface, used in conjunction with COM B
DO 15	DO1 5- way digital output	DO1 5 digital output interface, used in conjunction with COM B
DO 16	DO1 6- digit digital output	DO1 6 -bit digital output interface, used in conjunction with COM B

COM B	DO's COM side	DO 9 - DO 16 share a common COM B for use.
COM B	DO's COM side	DO 9 - DO 16 share a common COM B for use.
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-XXAX00A0G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
DO1	DO1 output indicator light	Green LED light; On: DO1 relay closed; Off: DO1 relay open.
DO2	DO2 output indicator light	Green LED light; On: DO 2 relay closed; Off: DO 2 relay open.
DO3	DO3 output indicator light	Green LED light; On: DO 3 relay closed; Off: DO 3 relay open.
DO4	DO4 output indicator light	Green LED light; On: DO 4 relay closed; Off: DO 4 relay open.
DO5	DO5 output indicator light	Green LED light; On: DO 5 relay closed; Off: DO 5 relay open.
DO6	DO6 output indicator light	Green LED light; On: DO 6 relay closed; Off: DO 6 relay open.
DO7	DO7 output indicator light	Green LED light; On: DO 7 relay closed; Off: DO 7 relay open.

DO8	DO8 output indicator light	Green LED light; On: DO 8 relay closed; Off: DO 8 relay open.
DO 9	DO 9 output indicator light	Green LED light; On: DO 9 relay closed; Off: DO 9 relay open.
DO 10	DO 10 output indicator light	Green LED light; On: DO1 0 relay closed; Off: DO1 0 relay open.
DO 11	DO 11 output indicator	Green LED light; On: DO 1 1 relay closed; Off: DO 1 1 relay open.
DO 12	DO 12 output indicator	Green LED light; On: DO12 relay closed; Off: DO12 relay open.
DO 13	DO 13 Output Indicator	Green LED light; On: DO1 3 relay closed; Off: DO1 3 relay open.
DO 14	DO 14 Output Indicator	Green LED light; On: DO1 4 relay closed; Off: DO1 4 relay open.
DO 15	DO 15 Output Indicator	Green LED light; On: DO15 relay closed; Off: DO15 relay open.
DO 16	DO 16 Output Indicator	Green LED light; On: DO1 6 relay closed; Off: DO1 6 relay open.

2.3.7 M31-XXEX00A0G-U-PN

M31-XXEX00A0G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DO1	DO1 digital output	DO1 digital output interface, used in conjunction with COMA
D O 2	DO 2 digital output	DO 2 is a digital output interface used in conjunction with COMA.
DO3	DO 3 digital output	DO 3 digital output interface, used in conjunction with COMA
DO4	DO 4 digital output	DO 4 digital output interface, used in conjunction with COMA
DO5	DO 5 digital output	DO 5 digital output interface, used in conjunction with COMA
DO 6	DO 6 digital output	DO 6 digital output interface, used in conjunction with COMA
DO7	DO 7 digital output	DO 7 digital output interface, used in conjunction with COMA
DO 8	DO 8 digital output	DO 8 digital output interface, used in conjunction with COMA
COM A	DO's COM side	DO1 - DO8 share COM A common terminal
DO 9	DO 9 digital output	DO 9 digital output interface, used in conjunction with COM 8
DO 10	DO 10 digital output	DO 10 digital output interface, used in conjunction with COM B
DO 11	DO1 1- way digital output	DO1 is a digital output interface for use with COM B.
DO 12	DO1 2 digital output	DO1 2 -bit digital output interface, used in conjunction with COM 8.
DO 13	DO1 3- way digital output	DO1 3 digital output interface, used in conjunction with COM B
DO 14	DO1 4- way digital output	DO1 4 -way digital output interface, used in conjunction with COM B
DO 15	DO1 5- way digital output	DO1 5 digital output interface, used in conjunction with COM B
DO 16	DO1 6- digit digital output	DO1 6 -bit digital output interface, used in conjunction with COM B

COM B	DO's COM side	DO 9 - DO 16 share a common COM B for use.
COM B	DO's COM side	DO 9 - DO 16 share a common COM B for use.
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-XXEX00A0G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flashing: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
DO1	DO1 output indicator light	Green LED light; On: DO1 relay closed; Off: DO1 relay open.
DO2	DO2 output indicator light	Green LED light; On: DO 2 relay closed; Off: DO 2 relay open.
DO3	DO3 output indicator light	Green LED light; On: DO 3 relay closed; Off: DO 3 relay open.
DO4	DO4 output indicator light	Green LED light; On: DO 4 relay closed; Off: DO 4 relay open.
DO5	DO5 output indicator light	Green LED light; On: DO 5 relay closed; Off: DO 5 relay open.
DO6	DO6 output indicator light	Green LED light; On: DO 6 relay closed; Off: DO 6 relay open.
DO7	DO7 output indicator light	Green LED light; On: DO 7 relay closed; Off: DO 7 relay open.

DO8	DO8 output indicator light	Green LED light; On: DO 8 relay closed; Off: DO 8 relay open.
DO 9	DO 9 output indicator light	Green LED light; On: DO 9 relay closed; Off: DO 9 relay open.
DO 10	DO 10 output indicator light	Green LED light; On: DO1 0 relay closed; Off: DO1 0 relay open.
DO 11	DO 11 output indicator	Green LED light; On: DO 1 1 relay closed; Off: DO 1 1 relay open.
DO 12	DO 12 output indicator	Green LED light; On: DO12 relay closed; Off: DO12 relay open.
DO 13	DO 13 Output Indicator	Green LED light; On: DO1 3 relay closed; Off: DO1 3 relay open.
DO 14	DO 14 Output Indicator	Green LED light; On: DO1 4 relay closed; Off: DO1 4 relay open.
DO 15	DO 15 Output Indicator	Green LED light; On: DO15 relay closed; Off: DO15 relay open.
DO 16	DO 16 Output Indicator	Green LED light; On: DO1 6 relay closed; Off: DO1 6 relay open.

2.3.8 M31-AXAX4040G-U-PN

M31-AXAX4040G-U-PN Port and Button Description:		
silkscreen	name	illustrate
DI 1	DI1 digital input	DI 1 is a digital input interface used in conjunction with COM 1 .
COM 1	DI 1 Common terminal for digital input	the DI1 digital output interface
DI 2	DI 2 digital input	DI 2 is a digital input interface used in conjunction with COM 2.
COM 2	DI 2 Common Terminal for Digital Input	the DI2 digital output interface
DI 3	DI 3 digital input	DI 3 is a digital input interface used in conjunction with COM 3 .
COM 3	DI 3 digital input common terminal	the DI3 digital output interface
DI 4	DI 4 digital input	DI 4 is a digital input interface used in conjunction with COM 4 .
COM 4	DI 4 Common Terminal for Digital Inputs	the DI4 digital output interface
DO 1	DO 1 digital output	DO 1 is a digital output interface used in conjunction with COM 1 .
COM 1	COM port of DO 1	DO 1 common terminal of digital output interface
DO 2	DO 2 digital output	DO 2 is a digital output interface used in conjunction with COM 2.
COM 2	COM port of DO 2	DO 2 common terminal of digital output interface
DO 3	DO 3 digital output	DO 3 is a digital output interface used in conjunction with COM 3 .

COM 3	DO 3 COM port	DO 3 digital output interface common terminal
DO 4	DO 4 digital output	DO 4 is a digital output interface used in conjunction with COM 4 .
COM 5	COM port of DO 4	DO 4 digital output interface common terminal
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-AXAX4040G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
DI1	DI1 Input Indicator	Green LED light; On: DI1 valid input; Off: DI1 invalid input.
DI2	DI2 Input Indicator	Green LED light; On: DI2 input is valid; Off: DI2 input is invalid.
DI3	DI3 Input Indicator	Green LED light; On: DI3 input is valid; Off: DI3 input is invalid.
DI4	DI4 Input Indicator	Green LED light; On: DI4 input is valid; Off: DI4 input is invalid.
DO1	DO1 output indicator light	Green LED light; On: DO1 relay closed; Off: DO1 relay open.
DO2	DO2 output indicator light	Green LED light; On: DO2 relay closed; Off: DO2 relay open.
DO3	DO3 output indicator	Green LED light; On: DO3 relay closed; Off: DO3 relay open.

	light	
DO4	DO4 output indicator light	Green LED light; On: DO4 relay closed; Off: DO4 relay open.

2.3.9 M31-XFXX0800G-U-PN

M31-XFXX0800G-U-PN Port and Button Description:		
silkscreen	name	illustrate
AI1+	AI1 Analog Current Input +	AI1 analog input + interface, used in conjunction with AI1-
AI1-	AI1 Analog Current Input -	AI1 Analog Input Interface, for use with AI1+
AI2+	AI 2 Analog Current Input +	AI 2 analog input + interface, used in conjunction with AI 2 -
AI2-	AI 2 Analog Current Input -	AI 2 Analog Input Interface, used in conjunction with AI 2+
AI3+	AI 3 Analog Current Input +	AI 3 analog input + interface, used in conjunction with AI 3 .
AI3-	AI 3 Analog Current Input -	AI 3 Analog Input Interface, for use with AI 3+
AI4+	AI 4 Analog Current Input +	AI 4 analog input + interface, used in conjunction with AI 4 .
AI4-	AI 4 Analog Current Input -	AI 4 Analog Input Interface, for use with AI 4+
AI5+	AI 5 Analog Current Input +	AI 5 analog input + interface, used in conjunction with AI 5 .
AI5-	AI 5 Analog Current Input -	AI 5 Analog Input Interface, for use with AI 5+
AI6+	AI 6 Analog Current Input +	AI 6 analog input + interface, used in conjunction with AI 6 .
AI6-	AI 6 Analog Current Input -	AI 6 Analog Input Interface, for use with AI 6+
AI7+	AI 7 Analog Current Input +	AI 7 analog input + interface, used in conjunction with AI 7 .
AI7-	AI 7 Analog Current Input -	AI 7 Analog Input Interface, for use with AI 7+
AI8+	AI 8 Analog Current Input +	AI 8 analog input + interface, used in conjunction with AI 8 .
AI8-	AI 8 Analog Current Input -	AI 8 Analog Input Interface, for use with AI 8+
NC	Empty terminal (Not Care)	No actual circuit connection function

NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings; Double-clicking within 2 seconds will automatically negotiate the I/O expansion module ;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-XFXX0800G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
AI 1	AI 1 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 2	AI 2 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 3	AI 3 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 4	A I4 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI5	AI5 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.

AI6	AI6 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI7	AI7 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI8	AI8 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.

2.3.10 M31-XAXX0A00G-U-PN

M31-XAXX0A00G-U-PN Port and Button Description:		
silkscreen	name	illustrate
AI1	AI1 Analog Current Input	AI1 analog input interface, used in conjunction with COM A
AI2	AI2 analog current input	AI2 analog input interface, used in conjunction with COM A
AI3	AI3 Analog Current Input	AI3 analog input interface, used in conjunction with COM A
AI4	AI4 Analog Current Input	AI4 analog input interface, used in conjunction with COM A
AI5	AI5 Analog Current Input	AI5 analog input interface, used in conjunction with COM A
AI6	AI6 Analog Current Input	AI6 analog input interface, used in conjunction with COM A
AI7	AI7 Analog Current Input	AI7 analog input interface, used in conjunction with COM A
AI8	AI8 Analog Current Input	AI8 analog input interface, used in conjunction with COM A
COMA	AI analog input public terminal	AI1 - AI 8 Shared COM A Common Terminal
AI9	AI9 Analog Current Input	AI9 analog input interface, used in conjunction with COM B
AI10	AI10 Analog Current Input	AI10 analog input interface, used in conjunction with COM B
AI11	AI11 Analog Current Input	AI11 analog input interface, used in conjunction with COM B
AI12	AI12 Analog Current Input	AI12 analog input interface, used in conjunction with COM B
AI13	AI13 Analog Current Input	AI13 analog input interface, used in conjunction with COM B
AI14	AI14 Analog Current Input	AI14 analog input interface, used in conjunction with COM B
AI15	AI15 Analog Current Input	AI15 analog input interface, used in conjunction with COM B

AI16	AI16 Analog Current Input	AI16 analog input interface, used in conjunction with COM B
COMB	AI analog input public terminal	AI9 - AI16 Shared COM B Common Terminal
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings; Double-clicking within 2 seconds will automatically negotiate the I/O expansion module ;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-XAXX0A00G-U-PN Indicator Light Description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
AI 1	AI 1 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 2	AI 2 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 3	AI 3 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 4	A I4 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.

AI5	AI5 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI6	AI6 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI7	AI7 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI8	AI8 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI9	AI9 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI10	AI10 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI11	AI11 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI12	AI 12 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI13	AI13 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI14	AI14 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI15	AI15 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI16	AI16 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.

2.3.11M31-XBXX0A00G-U-PN

M31-XBXX0A00G-U-PN Port and Button Description:		
silkscreen	name	illustrate
AI1	AI1 Analog Voltage Input	AI1 analog input interface, used in conjunction with COM A
AI2	AI2 analog voltage input	AI2 analog input interface, used in conjunction with COM A
AI3	AI3 Analog Voltage Input	AI3 analog input interface, used in conjunction with COM A
AI4	AI4 Analog Voltage Input	AI4 analog input interface, used in conjunction with COM A
AI5	AI5 Analog Voltage Input	AI5 analog input interface, used in conjunction with COM A
AI6	AI6 Analog Voltage Input	AI6 analog input interface, used in conjunction with COM A
AI7	AI7 Analog Voltage Input	AI7 analog input interface, used in conjunction with COM A

AI8	AI8 Analog Voltage Input	AI8 analog input interface, used in conjunction with COM A
COMA	AI analog input public terminal	AI1 - AI 8 Shared COM A Common Terminal
AI9	AI9 analog voltage input	AI9 analog input interface, used in conjunction with COM B
AI10	AI10 analog voltage input	AI10 analog input interface, used in conjunction with COM B
AI11	AI11 Analog Voltage Input	AI11 analog input interface, used in conjunction with COM B
AI12	AI12 analog voltage input	AI12 analog input interface, used in conjunction with COM B
AI13	AI13 Analog Voltage Input	AI13 analog input interface, used in conjunction with COM B
AI14	AI14 Analog Voltage Input	AI14 analog input interface, used in conjunction with COM B
AI15	AI15 Analog Voltage Input	AI15 analog input interface, used in conjunction with COM B
AI16	AI16 Analog Voltage Input	AI16 analog input interface, used in conjunction with COM B
COMB	AI analog input public terminal	AI9 - AI16 Shared COM B Common Terminal
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings; Double-clicking within 2 seconds will automatically negotiate the I/O expansion module ;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-XBXX0A00G-U-PN Indicator Light Description:

silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively

		searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flashing: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
AI 1	AI 1 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 2	AI 2 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 3	AI 3 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI 4	A I4 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI5	AI5 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI6	AI6 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI7	AI7 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI8	AI8 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI9	AI9 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI10	AI10 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI11	AI11 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI12	AI 12 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI13	AI13 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI14	AI14 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI15	AI15 Input Indicator Light	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.
AI16	AI16 Input Indicator	Green LED light; On: Normal input reaches 1% or more of the range; Off: Not effectively connected; Rapid flashing: Exceeds the range by 10%.

2.3.12 M31-XXXA0008G-U-PN

M31-XXXA0008G-U-PN Port and Button Description:		
silkscreen	name	illustrate
AO1	AO1 Analog Current Output (Positive)	AO1 analog output (positive) interface, used in conjunction with GND1.
GND1	AO1 Analog Current Output (Negative Terminal)	AO1 analog output (negative terminal) interface
AO2	AO 2 Analog Current Output (Positive)	AO 2 analog output (positive) interface, used in conjunction with GND 2.
GND2	AO 2 Analog Current Output (Negative Terminal)	AO 2 analog output (negative) interface
AO3	AO 3 Analog Current Output (Positive)	AO 3 analog output (positive) interface, used in conjunction with GND 3.
GND3	AO 3 Analog Current Output (Negative Terminal)	AO 3 analog output (negative) interface
AO4	AO 4 Analog Current Output (Positive)	AO 4 analog output (positive) interface, used in conjunction with GND 4.
GND4	AO 4 Analog Current Output (Negative)	AO 4 Analog Output (Negative) Interface
AO5	AO 5 Analog Current Output (Positive)	AO 5 analog output (positive) interface, used in conjunction with GND 5.
GND5	AO 5 Analog Current Output (Negative Terminal)	AO 5 Analog Output (Negative) Interface
AO6	AO 6 Analog Current Output (Positive)	AO 6 analog output (positive) interface, used in conjunction with GND 6.
GND6	AO 6 Analog Current Output (Negative)	AO 6 Analog Output (Negative) Interface
AO7	AO 7 Analog Current Output (Positive)	AO 7 analog output (positive) interface, used in conjunction with GND 7.
GND7	AO 7 Analog Current Output (Negative)	AO 7 Analog Output (Negative) Interface
AO8	AO 8 Analog Current Output (Positive)	AO 8 analog output (positive) interface, used in conjunction with GND 8.
GND8	AO 8 Analog Current Output (Negative)	AO 8 Analog Output (Negative) Interface
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings; Double-clicking within 2 seconds will automatically negotiate the I/O expansion module ;

A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

M31-XXXA0008G-U-PN indicator light description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
AO 1	AO 1 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO 2	AO 2 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO 3	AO 3 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO 4	AO 4 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO5	AO5 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO6	AO6 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO7	AO7 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO8	AO8 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.

2.3.13M31-XXXB0008G-U-PN

M31-XXXB0008G-U-PN Port and Button Description:		
silkscreen	name	illustrate
AO1	AO1 Analog Voltage Output (Positive)	AO1 analog output (positive) interface, used in conjunction with GND1.
GND1	AO1 Analog Voltage Output (Negative Terminal)	AO1 analog output (negative terminal) interface
AO2	AO2 analog voltage output (positive terminal)	AO2 analog output (positive) interface, used in conjunction with GND2.
GND2	AO2 analog voltage output (negative terminal)	AO2 analog output (negative terminal) interface
AO3	AO3 analog voltage output (positive terminal)	AO3 analog output (positive) interface, used in conjunction with GND3.
GND3	AO3 analog voltage output (negative terminal)	AO3 analog output (negative terminal) interface
AO4	AO4 Analog Voltage Output (Positive)	AO4 analog output (positive) interface, used in conjunction with GND4.
GND4	AO4 analog voltage output (negative terminal)	AO4 analog output (negative terminal) interface
AO5	AO5 analog voltage output (positive terminal)	AO5 analog output (positive) interface, used in conjunction with GND5.
GND5	AO5 analog voltage output (negative terminal)	AO5 analog output (negative terminal) interface
AO6	AO6 analog voltage output (positive terminal)	AO6 analog output (positive) interface, used in conjunction with GND6.
GND6	AO6 Analog Voltage Output (Negative Terminal)	AO6 analog output (negative) interface
AO7	AO7 Analog Voltage Output (Positive)	AO7 analog output (positive) interface, used in conjunction with GND7.
GND7	AO7 Analog Voltage Output (Negative)	AO7 Analog Output (Negative) Interface
AO8	AO8 analog voltage output (positive terminal)	AO8 analog output (positive) interface, used in conjunction with GND8.
GND8	AO8 analog voltage output (negative terminal)	AO8 analog output (negative terminal) interface
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings; Double-clicking within 2 seconds will automatically negotiate the I/O expansion module;

A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN2	network port	Standard RJ45 network cable interface

M31-XXXB0008G-U-PN indicator light description:		
silkscreen	name	illustrate
RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
AO 1	AO 1 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO 2	AO 2 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO 3	AO 3 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO 4	AO 4 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO5	AO5 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO6	AO6 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO7	AO7 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.
AO8	AO8 output indicator light	Green LED light; On: Normal output reaches 1% or more of the range; Off: No effective output.

2.3.14M31 -X D XX0 6 00G-U -PN

M31- X D XX0 6 00 GU -PN Port and Button Description:		
silkscreen	name	illustrate
RTD1+	1-channel PT100 input	1-channel PT100 input + interface
RTD1-	1-channel PT100 input	- channel PT100 input interface
RTD1-	1-channel PT100 input	- channel PT100 input interface
RTD2+	2-channel PT100 input	2-channel PT100 input + interface
RTD2-	2-channel PT100 input	- channel PT100 input interface
RTD2-	2-channel PT100 input	- channel PT100 input interface
RTD3+	3-channel PT100 input	3-channel PT100 input + interface
RTD3-	3-channel PT100 input	- channel PT100 input interface
RTD3-	3-channel PT100 input	- channel PT100 input interface
RTD4+	4-channel PT100 input	4-channel PT100 input + interface
RTD4-	4-channel PT100 input	- channel PT100 input interface
RTD4-	4-channel PT100 input	- channel PT100 input interface
RTD5+	5-channel PT100 input	5-channel PT100 input + interface
RTD5-	5-channel PT100 input	- channel PT100 input interface
RTD5-	5-channel PT100 input	- channel PT100 input interface
RTD6+	6-channel PT100 input	6-channel PT100 input + interface
RTD6-	6-channel PT100 input	- channel PT100 input interface
RTD6-	6-channel PT100 input	- channel PT100 input interface
NC	Empty terminal (Not Care)	No actual circuit connection function
Reload	Factory reset/auto-negotiation	Press and hold for 5-10 seconds to restore factory settings; Double-clicking within 2 seconds will automatically negotiate the I/O expansion module ;
A(RS485)	RS485 A interface	RS485 A interface
B(RS485)	RS485 B interface	RS485 B interface
G(RS485)	RS485 G interface	RS485 G interface
PE	Grounding	Grounding
V- (DC9-36V)	negative terminal of power supply	DC (9-36V) power supply negative terminal interface
V+ (DC9-36V)	Positive power supply	DC (9-36V) power supply positive terminal interface
LAN1	network port	Standard RJ45 network cable interface
LAN 2	network port	Standard RJ45 network cable interface

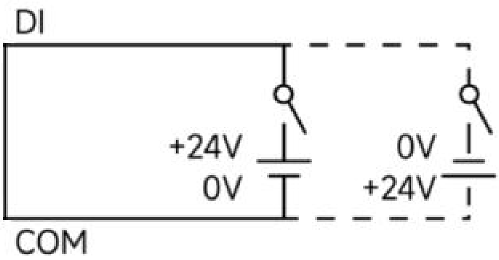
M31-XDXX0600G-U -PN Indicator Light Description:		
silkscreen	name	illustrate

RUN	Power indicator light	Red LED light; On: Connection to master station initialization completed; Off: Device malfunction; 1Hz flashing: Not connected to master station, initialization incomplete; 4Hz flashing: Configuration software actively searches for and requests device flashing; 20Hz flashing: Gateway serial port data transmission and reception;
BUS	Device bus status indicator	Yellow LED light; On: Internal bus of the device is operating normally; Off: Internal bus of the device is completely malfunctioning. Flickering: There may be an abnormality in the operation of the device's internal bus.
STA	Bus communication indicator	Blue LED light; flashing: indicates data exchange between the host and the expansion module; off: indicates no data exchange between the host and the expansion module.
RTD1	RTD1 Input Indicator	Green LED light; On: Sensor is successfully connected; Off: Sensor is not successfully connected;
RTD2	RTD2 Input Indicator Light	Green LED light; On: Sensor is successfully connected; Off: Sensor is not successfully connected;
RTD3	RTD3 Input Indicator Light	Green LED light; On: Sensor is successfully connected; Off: Sensor is not successfully connected;
RTD4	RTD4 Input Indicator	Green LED light; On: Sensor is successfully connected; Off: Sensor is not successfully connected;
RTD5	RTD5 Input Indicator Light	Green LED light; On: Sensor is successfully connected; Off: Sensor is not successfully connected;
RTD6	RTD6 Input Indicator Light	Green LED light; On: Sensor is successfully connected; Off: Sensor is not successfully connected;

3 Wiring Instructions

3.1Device connection

3.1.1 DI connection

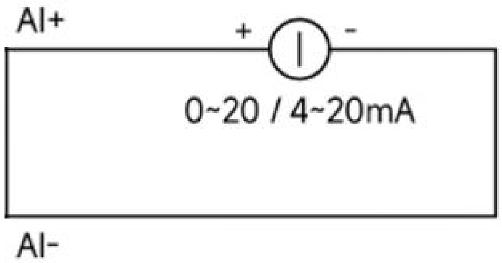


PNP input **NPN input**

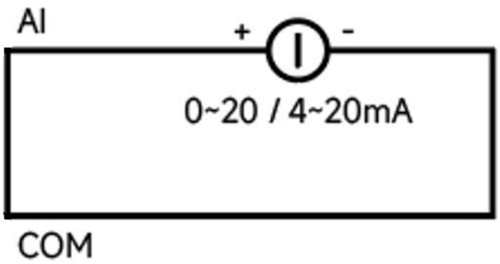
Note: DI is an NPN or PNP active input, and the voltage range supported is only 10 V to 28 V.

3.1.2 AI Connectivity

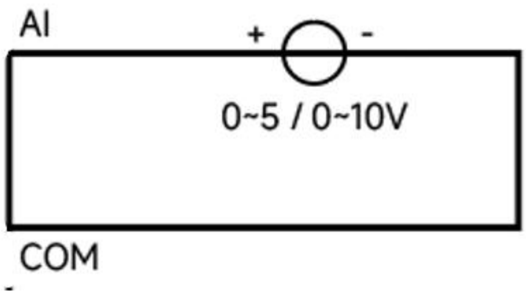
Differential analog current acquisition:



Single-ended analog current acquisition:

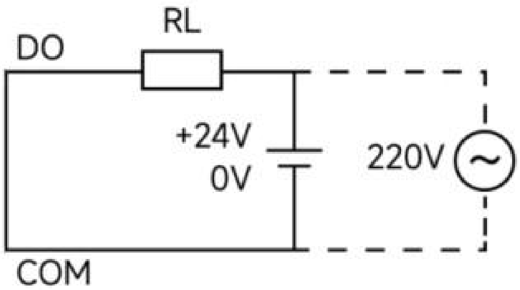


Single-ended analog voltage acquisition:



3.1.3 DO connection

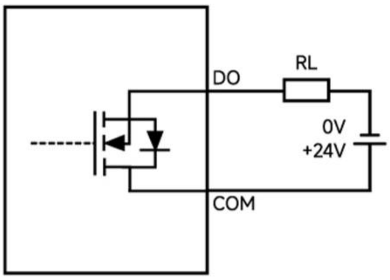
3.1.3.1 Relay type



Note: 1. A single relay can support a maximum of 5A.

2. The maximum total current supported for each group (same COM common terminal) is 8A.

3.1.3.2 transistor type

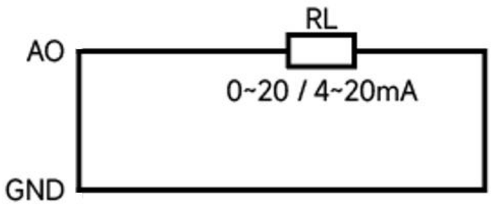


Note: 1. A single channel can support a maximum of 0.5A .

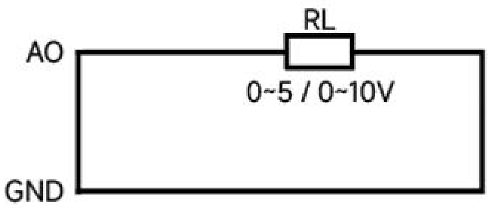
2. The maximum total current supported by each group (same COM common terminal) is 4 A , and the voltage range is 10-30V .

3.1.4 AO connection

Current type AO :

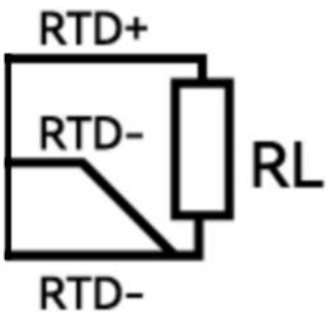


Voltage type AO :



3.1.5 PT100 connection

Three-wire PT100 RTD data acquisition:



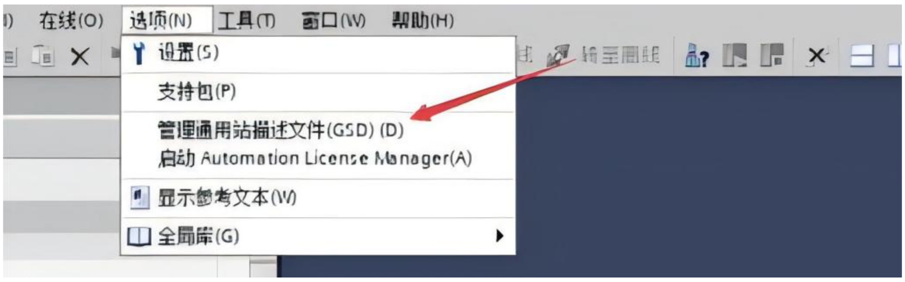
4 Tutorial on loading devices using TIA Portal

4.1 Preparation before connection

1. Prepare the necessary XML file as follows:
   GSDML-V2.3-EBYTE-P31-20251217xml

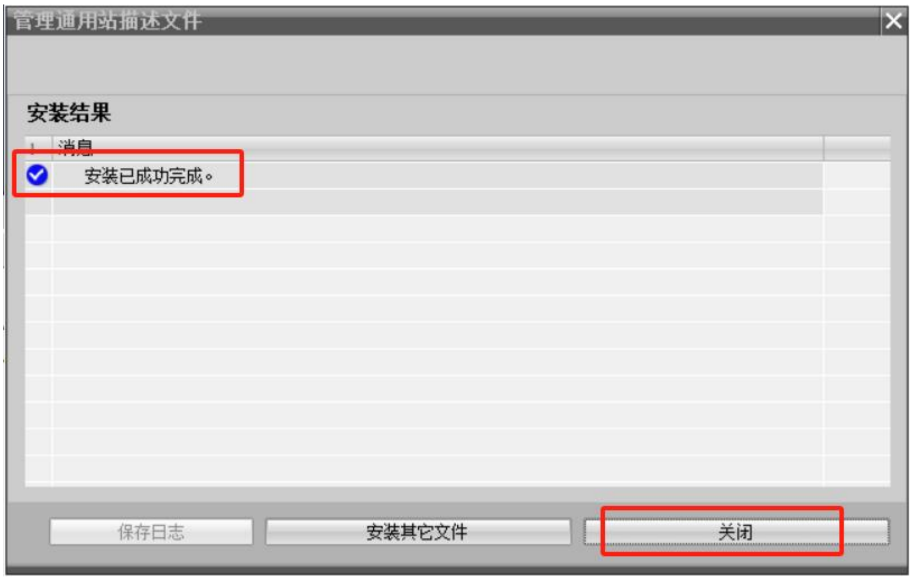
4.2 Add GSDML file to TIBRA

1. Open TIA Portal, click Options, and then Manage General Site Files (GSD).



2. Add the GSDML file. Locate the folder where the GSD is stored, select the folder, and the software will automatically scan for the GSD file. Then, select the GSD file and click Install.

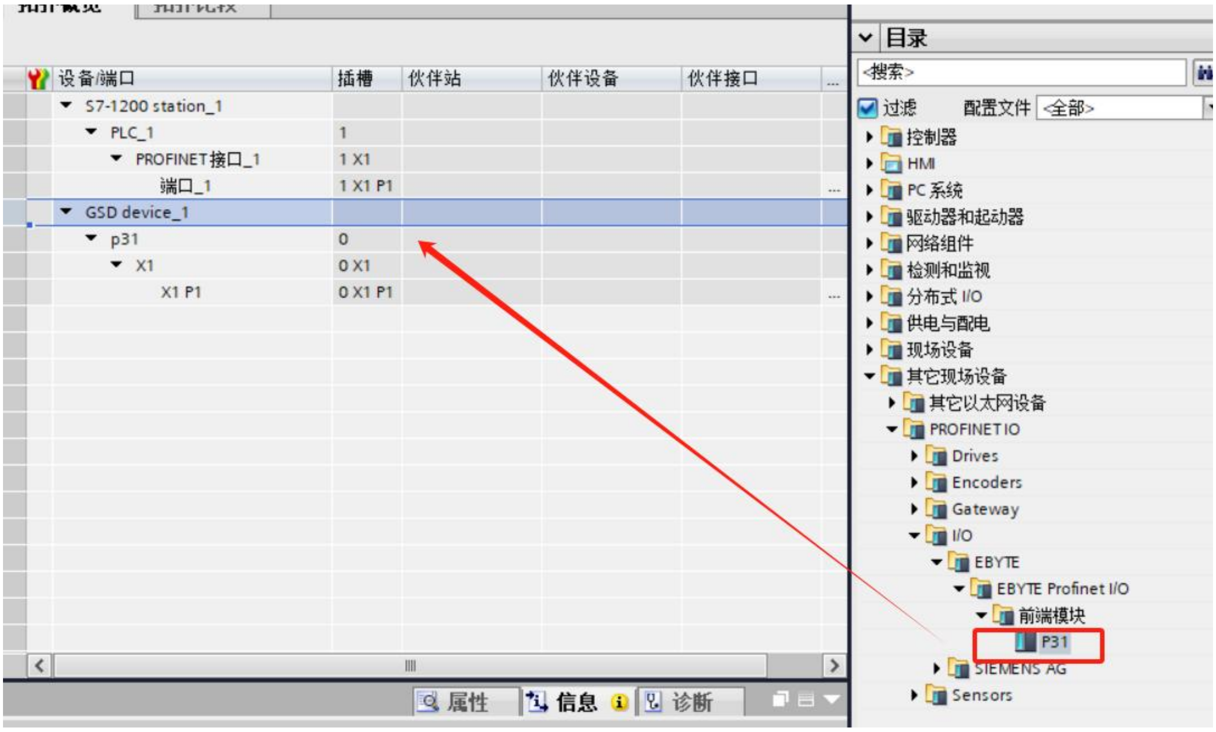
3. Once the installation is complete, simply click "Close," and then you can configure the device.



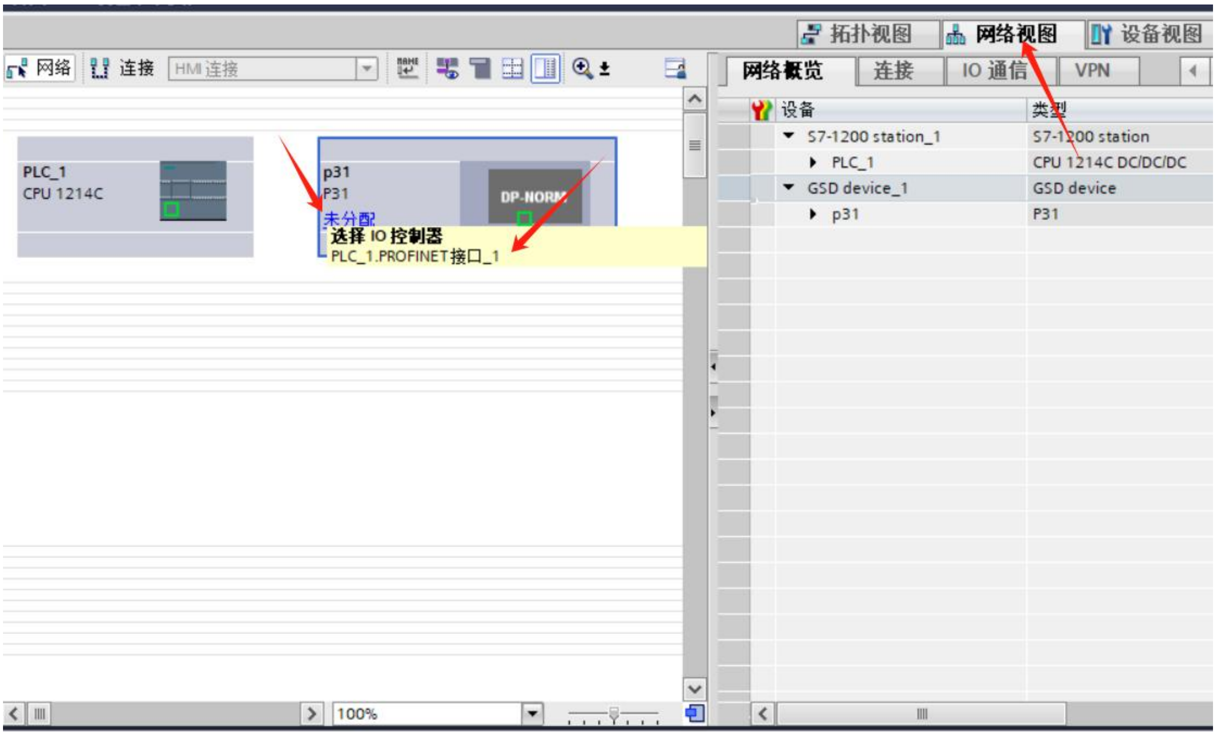
4.3 Add Profinet IO device to project

The following example demonstrates adding an M31-AFAX4440G-U-PN main unit and a GAFAX4440-U expansion module.

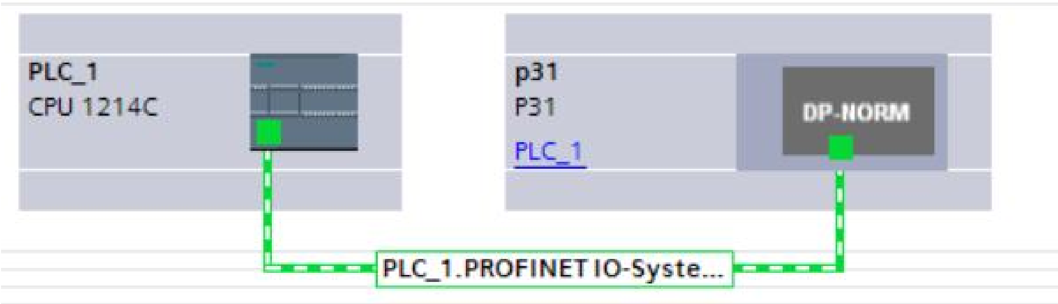
1. To create or open a new project, first add a controller device, then add the corresponding IO module in the device configuration interface. Double-click the module to add it successfully, as shown in the figure:



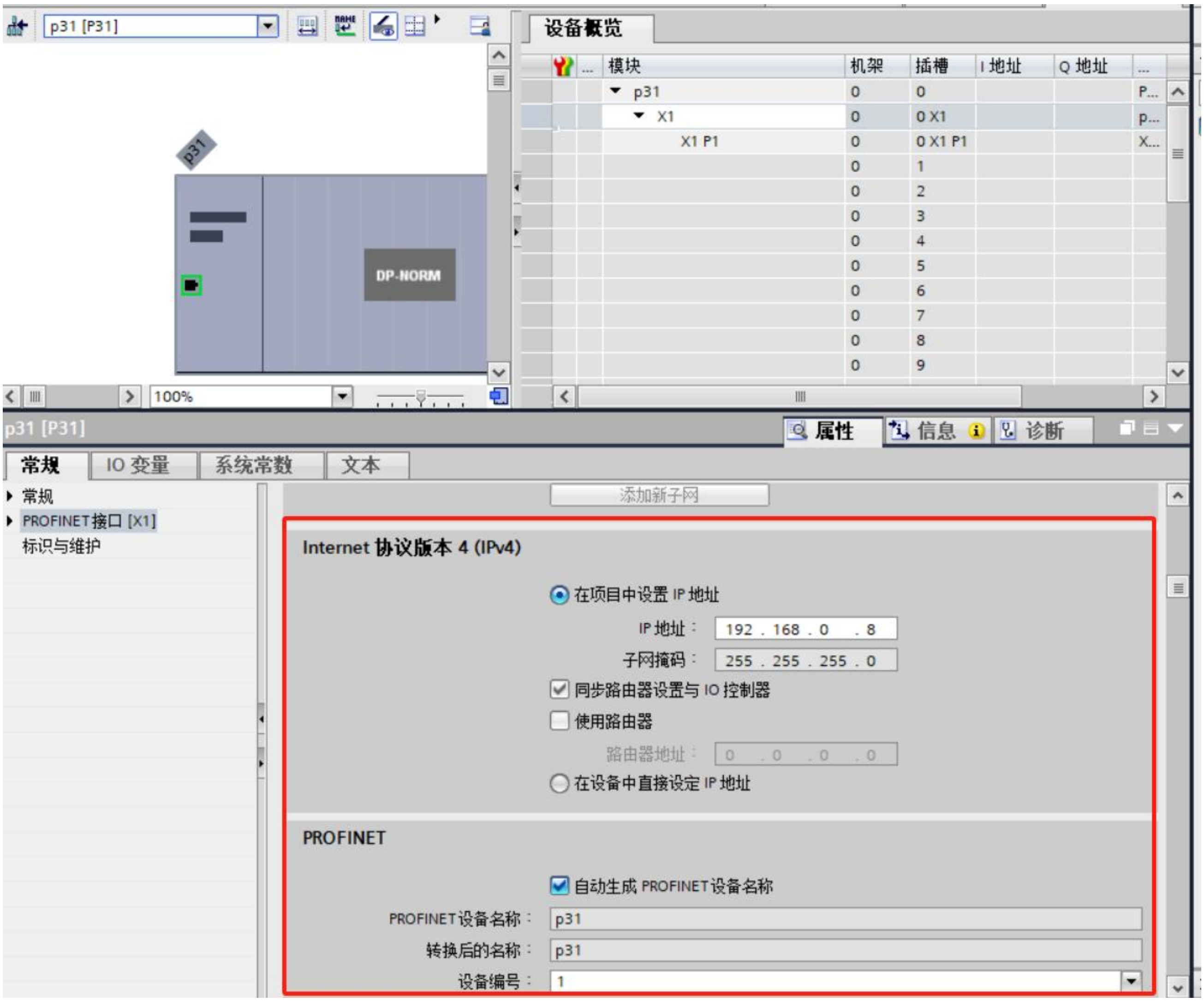
2. Configure devices in the network view.



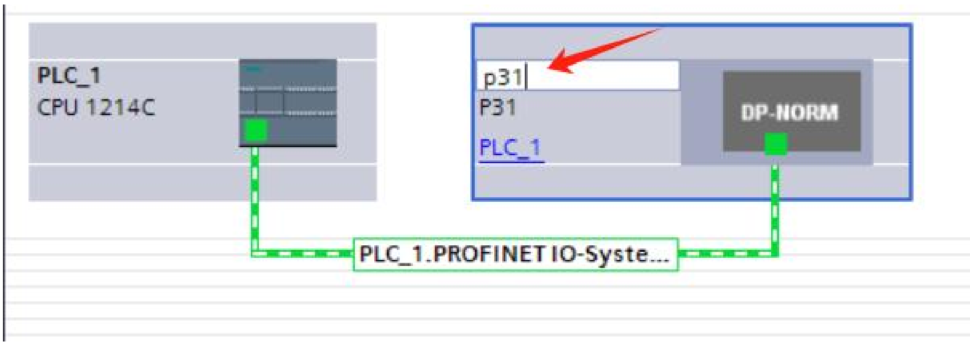
Once completed, it will look like this:



3. In the device view, first locate and select the newly added device, then double-click the module in the image. Next, in the general configuration interface, modify the IP address and device name, ensuring they match the module itself. Alternatively, you can choose to set the IP address directly on the device.



You can also click here to directly modify the device name.



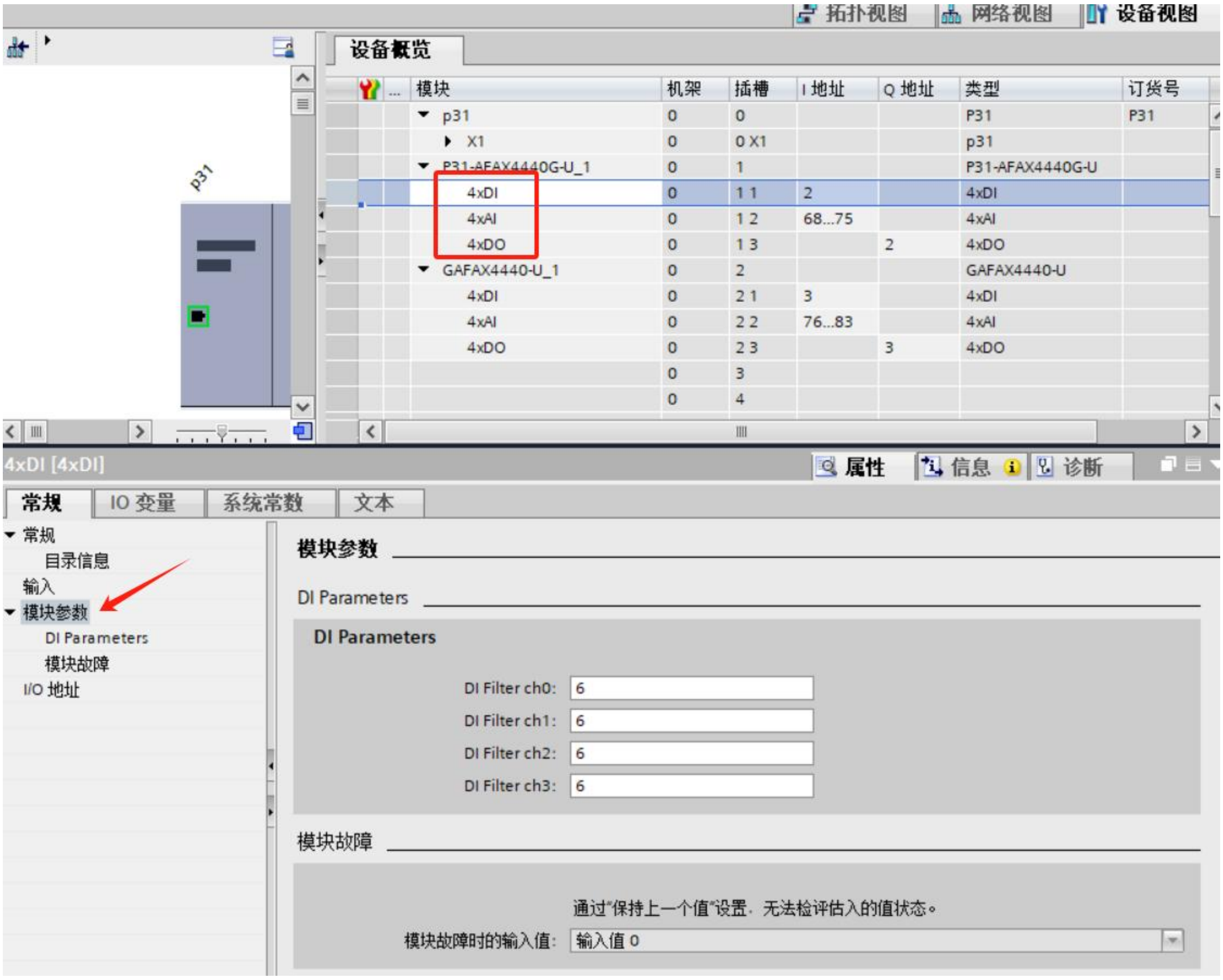
4. Select the p31 module, click Device View, and then add M31-AFAX4440G-U-PN and GAFAX4440-U according to the device splicing order.

4.4 Check the starting addresses of points I and Q.

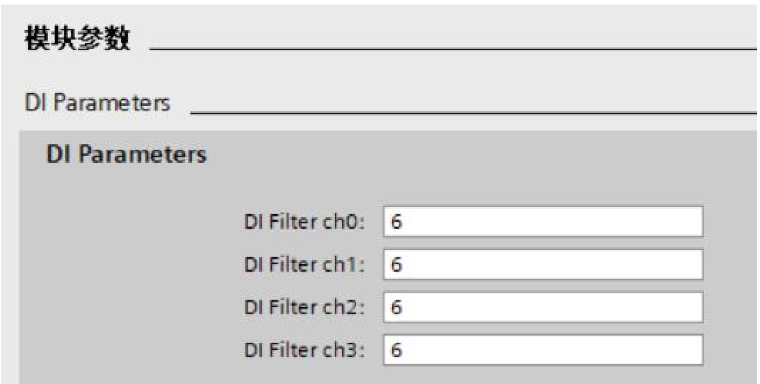
1. The device view on p31 shows the address and length allocated to each submodule:

4.5 Instructions for using the IO function:

1. Click on the corresponding IO type, then click on module parameters to see the relevant settings options for the IO function:

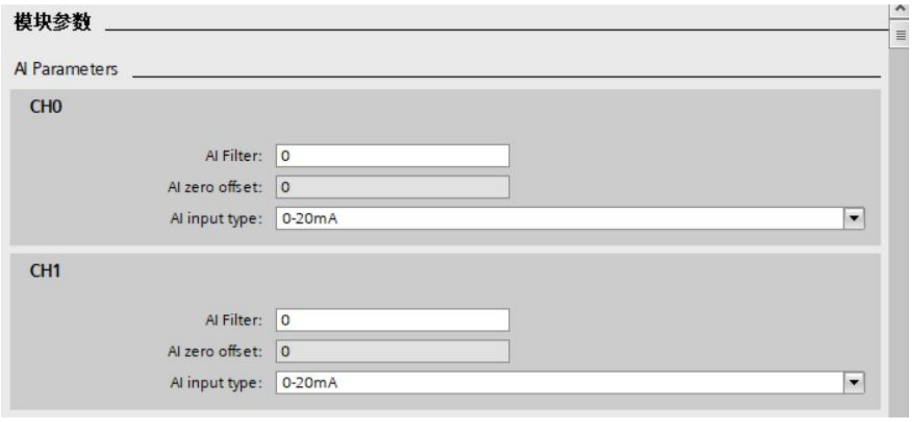


2. DI Parameters:



DI Filter: The DI filtering time for each channel can be set individually.

3. AI Parameters:

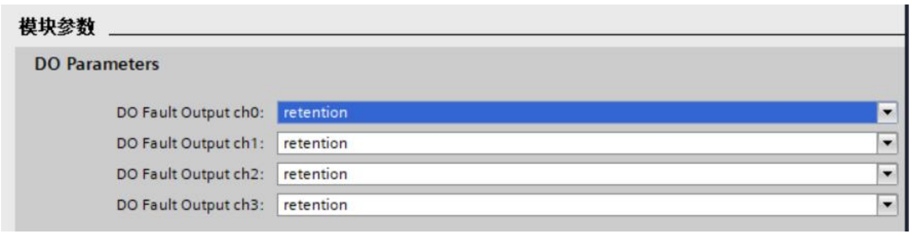


AI Filter: The AI filtering level for each channel can be set individually. The higher the level, the longer the filtering time.

AI zero offset: (Cannot be set)

AI input type (AI output type): Current type is available in ranges of 0~20mA, 4~20mA, and -20~20mA; Voltage type is available in ranges of 0~5V, 0~10V, -5~5V, and -10~10V.

4. DO Parameters:



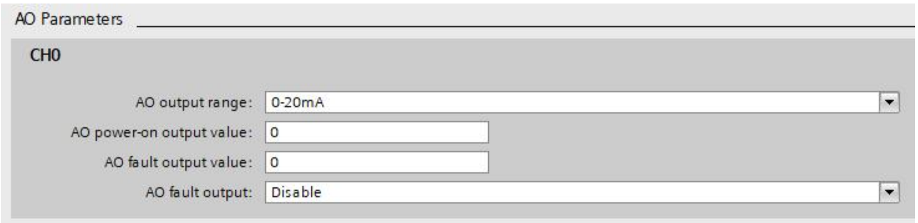
DO Fault Output: When the bus malfunctions, the output status of each channel's DO can be set individually.

Retention: In the event of a fault, retain the current state of the DO output.

Reset: In case of a fault, resets the DO output to its previous state.

Set (Set bit): In case of a fault, set the state before the DO output is set.

5. AO Parameters:



AO output range: Current type is available in 0~20mA and 4~20mA ranges; Voltage type is available in 0~5V and 0~10V ranges.

AO power-on output value: A preset fixed value that the AO channel automatically outputs after the device is powered on.

AO fault output value: The emergency output value that the AO channel automatically switches to when the device detects a communication interruption, bus abnormality, or module failure.

AO fault output: Controls the on/off state of the AO fault output function. When enabled, the AO channel will execute according to the preset "fault output value" when the device fails; when disabled, the AO channel will retain the last normal output value when a fault occurs.

6. PT100 Parameters:

Read the temperature data collected by PT100 and directly convert it to the actual temperature.

In Profinet IO communication, the temperature data of PT100 is in 16-bit signed integer (INT) format and is used as input data, which is uploaded in real time by the slave station to the input process image area of the master station.

Temperature conversion: The read INT value ranges from -2000 to 8500, and has a linear correspondence with the actual temperature (-2000 corresponds to -200℃, 8500 corresponds to 850℃). The conversion formula is:

That is: Actual temperature = Collected value / 10 (°C)

[Status Diagnostic Value Explanation] When the INT value read from the Profinet data area is one of the following special values, it indicates a fault state for the corresponding channel, rather than the actual temperature value:

32766: Sensor not connected or wired out (wired out when temperature exceeds 870℃)

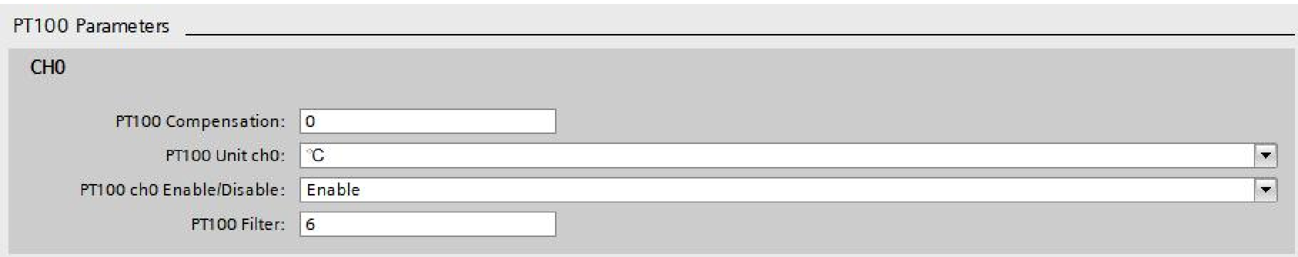
-32766: Short circuit condition (when the temperature is below -220℃, it is determined to be a short circuit condition)

32765: Chip fault (indicates that the ADC acquisition unit to which this channel belongs is malfunctioning and cannot complete the temperature signal conversion).

32767: Temperature overflow (triggered when the temperature exceeds 850°C but does not reach 870°C)

-32768: Temperature overflow (triggered when the temperature is below -200°C but not yet -220°C)

-32767: Channel is disabled



PT100 Compensation: Value range -200 to 200 (unit: 0.1℃, default value 0), used to compensate and calibrate the measured temperature of the channel. The compensation value is directly added to the final temperature result.

PT100 Unit ch0 (PT100 channel unit): Numerical range (°C; °F; K). After the setting takes effect, the integer and floating-point values of the project quantity will be automatically converted according to the selected unit.

PT100 ch0 Enable/Disable (PT100 channel disabled): When this channel is disabled , the engineering integer and floating-point registers for this channel will return the diagnostic value -32767.

PT100 Filter (PT100 filter parameters): Data range 1-16, default parameter 6 .

5 Tutorial on loading devices using STEP 7

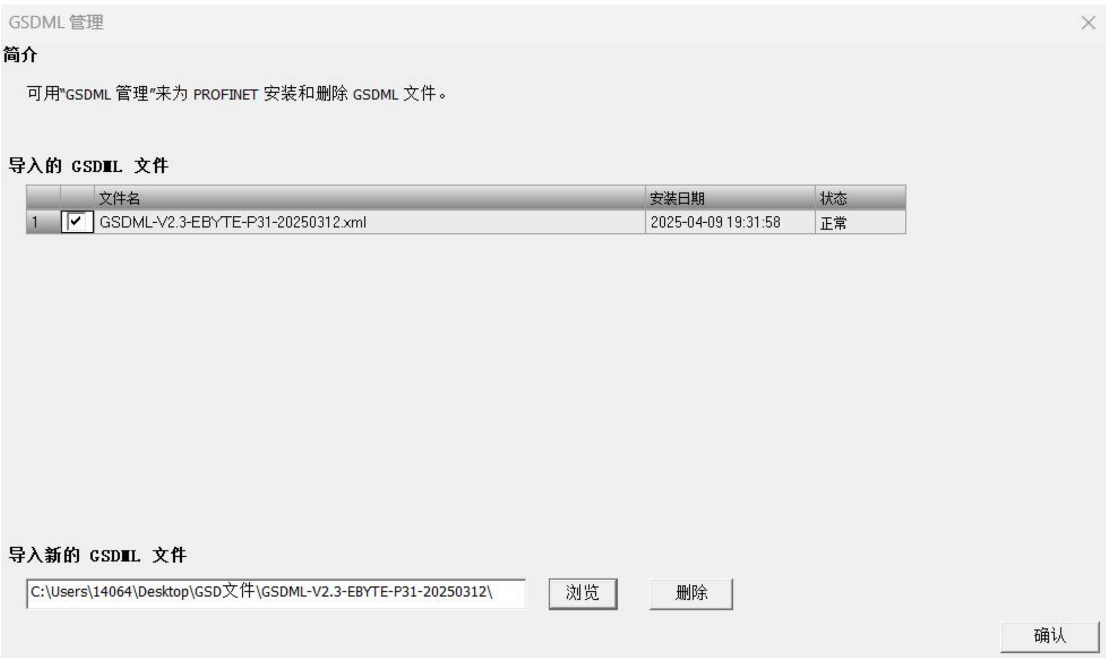
5.1 Preparation before connection

1. Prepare the necessary XML file as follows:
   GSDML-V2.3-EBYTE-P31-20251217xml

5.2 STEP 7 Add GSDML file

1. Under the File menu, access GSDML Management.

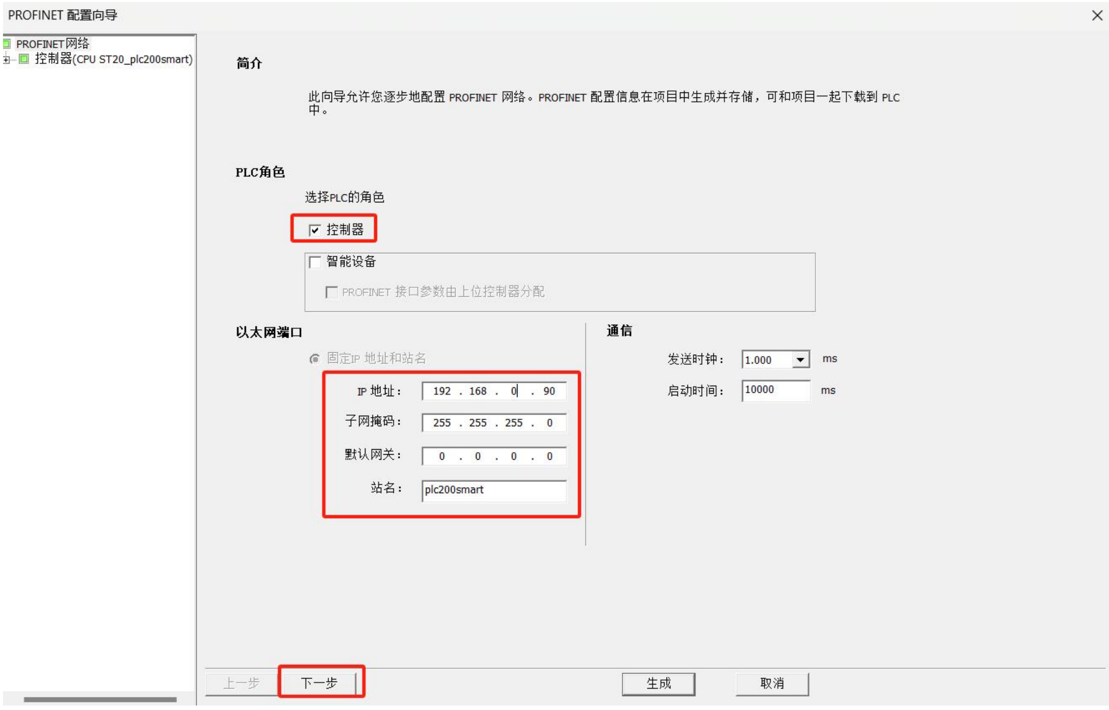
2. Add GSDML file



5.3 Add Profinet device to project

1. Select the PROFINET command under the Tools menu.

2. Select the PLC role as PLC controller, and set the corresponding PLC controller IP and other relevant parameters. Click Next when finished.

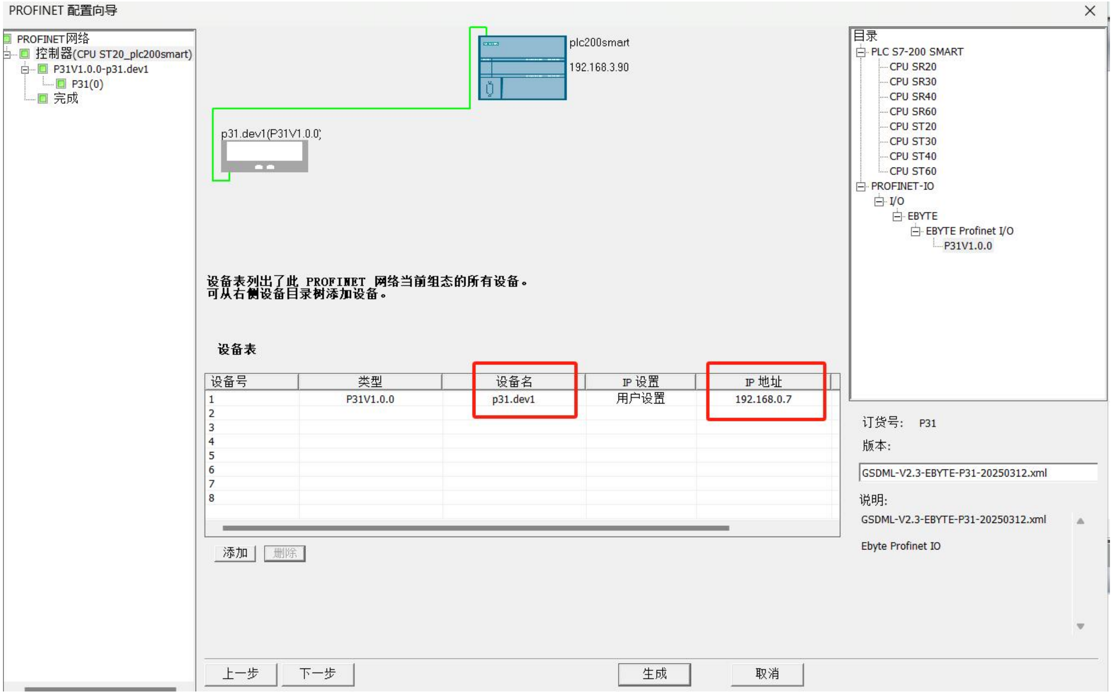


3. In the right-hand column, select PROFINET-IO→I/O→EBYTE→TION→EBYTE Profinet I/O→P31V1.0.0, then hold down the left mouse button and drag it into the table on the left.

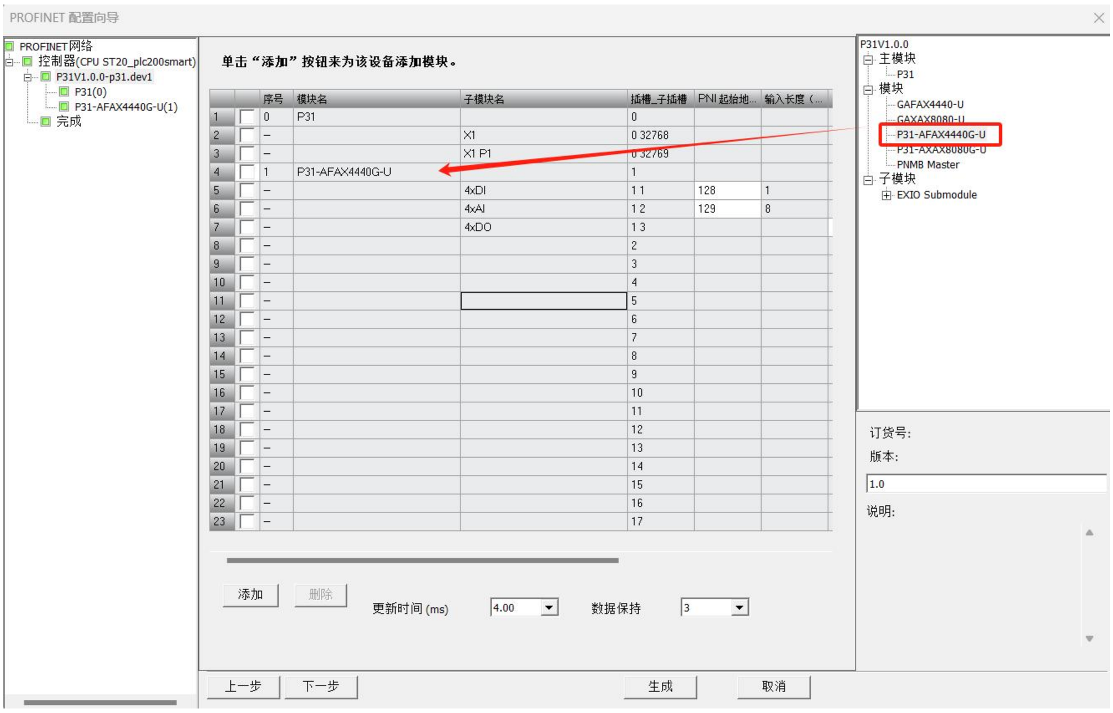
4. Double-click the device name field, enter the corresponding device name. There cannot be the same device

name in the same project. Similarly, set the IP address, keeping it in the same network segment as the PLC controller. Finally, click Generate.

Note: The device name set here must be consistent with the device. If you are unsure of the device name, you can set it arbitrarily first and then change the device name to match the device name later. The IP address set here will be used to set the IP address of devices with the same device name during configuration.



5. After completing the relevant settings, click to enter the device view operation interface. In the device overview area, drag the corresponding device model name (M31-AFAX4440G-U-PN) into slot 1.



Note: If the P31 host has expansion I/O modules, the model names of the expansion modules must be dragged into the slots in sequence. Otherwise, the device will report a configuration error!

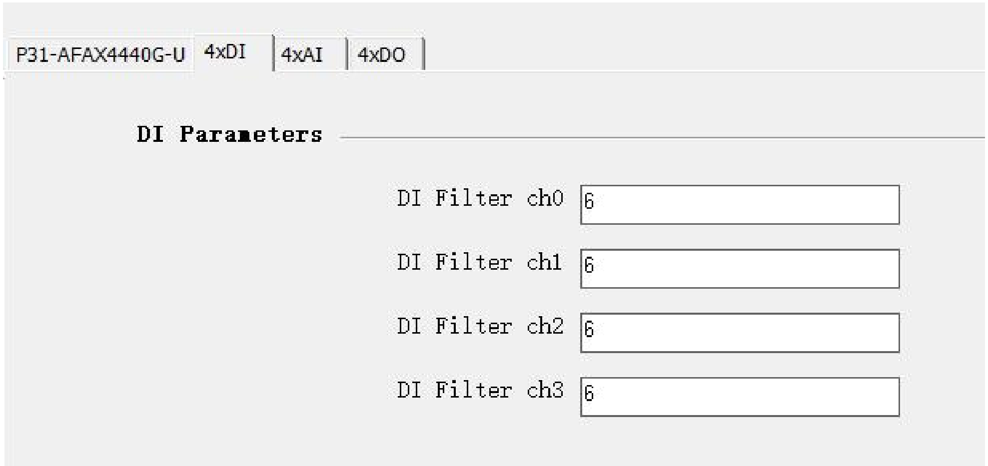
5.4 Check the starting addresses of points I and Q.

1. By sliding the bottom bar, you can see the starting address of DI, AI, and DO, as well as their length (in bytes).

5.5 Instructions for using the IO function:

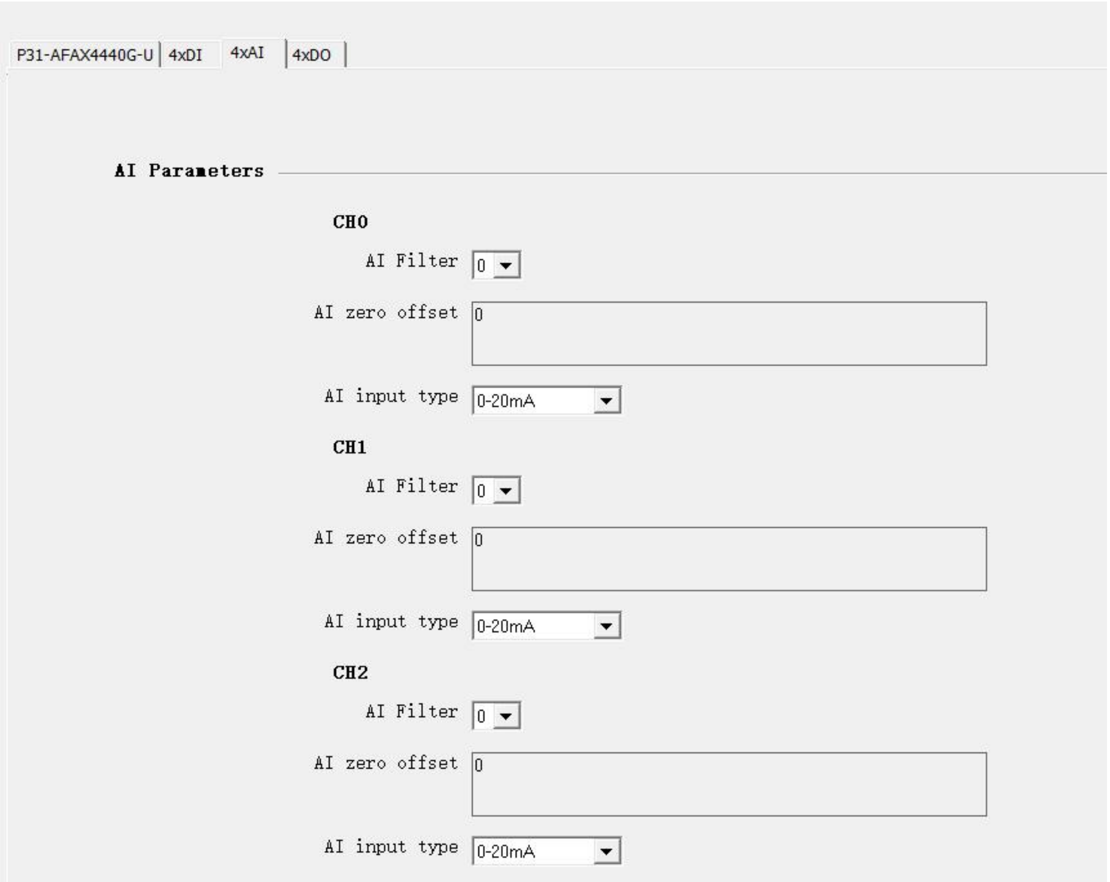
1. Select the device you just added (M31-AFAX4440G-U-PN). Here, you need to configure the corresponding DI, AI, and DO parameters according to the on-site usage requirements.

2. DI Parameters:



DI Filter: The DI filtering time for each channel can be set individually.

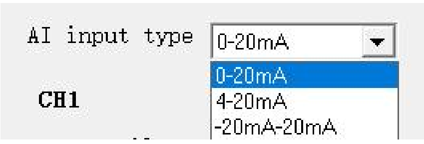
3. AI Parameters:



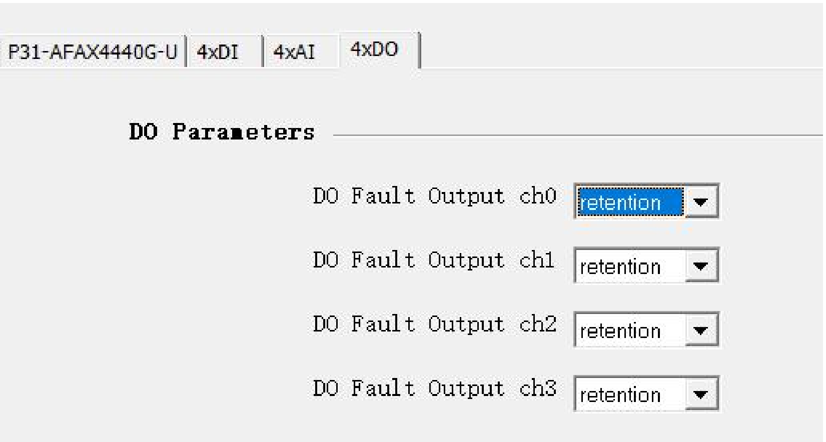
AI Filter: The AI filtering level for each channel can be set individually. The higher the level, the longer the filtering time.

AI zero offset: (Not configurable)

AI input type (AI output type): Selectable ranges are 0~20mA, 4~20mA, and -20~20mA.



4. DO Parameters:

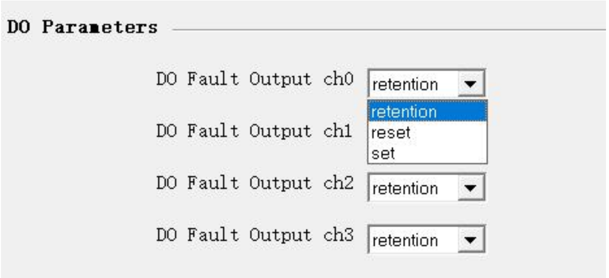


DO Fault Output: When the bus malfunctions, the output status of each channel's DO can be set individually.

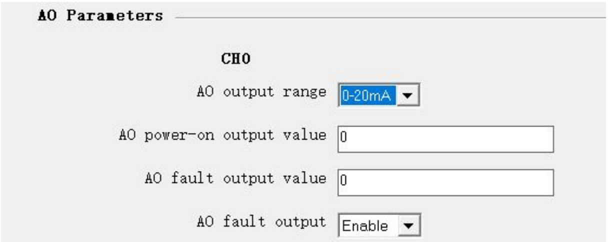
Retention: In the event of a fault, retain the current state of the DO output.

Reset: In case of a fault, resets the DO output to its previous state.

Set (Set): In case of a fault, set the state before the DO output is set.



5. AO Parameters:



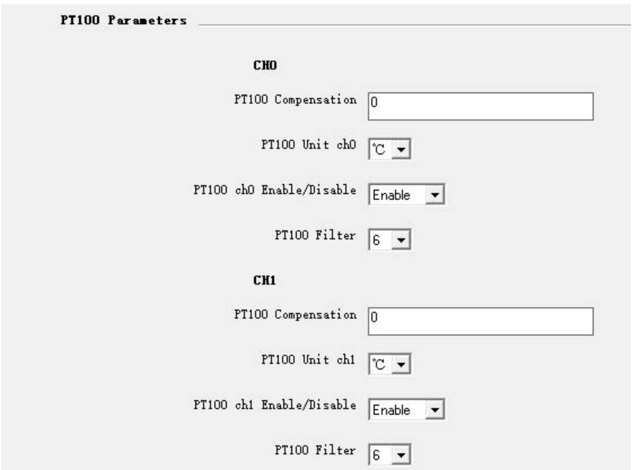
AO output range: Current type is available in 0~20mA and 4~20mA ranges; Voltage type is available in 0~5V and 0~10V ranges.

AO power-on output value: A preset fixed value that the AO channel automatically outputs after the device is powered on.

AO fault output value: The emergency output value that the AO channel automatically switches to when the device detects a communication interruption, bus abnormality, or module failure.

AO fault output: Controls the on/off state of the AO fault output function. When enabled, the AO channel will execute according to the preset "fault output value" when the device fails; when disabled, the AO channel will retain the last normal output value when a fault occurs.

6. PT100 Parameters:



PT100 Compensation: Value range -200 to 200 (unit: 0.1°C, default value 0), used to compensate and calibrate the measured temperature of the channel. The compensation value is directly added to the final temperature result.

PT100 Unit ch0 (PT100 channel unit): Numerical range (°C; °F; K). After the setting takes effect, the integer and floating-point values of the project quantity will be automatically converted according to the selected unit.

PT100 ch0 Enable/Disable (PT100 channel disabled): When this channel is disabled , the engineering integer and

floating-point registers for this channel will return the diagnostic value -32767.
PT100 Filter (PT100 filter parameters): Data range 1-16, default parameter 6 .

After configuring the relevant parameters, click "Generate".
At this moment, we have successfully completed the communication connection routine between Siemens PLC S7-200smart and Profinet distributed I/O (M31-AFAX4440G-U-PN).

6 Product Function Introduction

6.1 IO point number expansion

Note: Do not operate the equipment while it is powered on during the assembly process, otherwise it may cause damage to the equipment!

This product supports the splicing of 15 expansion modules, and the host can synchronize all 15 IO expansion modules in as little as 1ms.

The M31-U-PN series distributed I/O host adopts an scalable architecture design, in which the I/O expansion module can be used to expand the M31-U-PN series host. Simply connect the I/O expansion module to the host slot and then slide down the latch to firmly connect the host and the I/O expansion module together.

The specific steps are as follows:

- First, ensure the host is not powered on and that the host's slider is in the UNLK position. Then, connect the IO expansion module to the host, as shown in the following figure:

- After connecting the IO expansion module, slide the host to the LOCK position and then power on the host .
- In TIA Portal/STEP 7, simply drag and drop the corresponding module model according to the module assembly order.

6.2 Profinet to Modbus RTU gateway function

1. This product provides one RS485 interface and supports eight command nodes.
2. It can be used as a Modbus RTU master station .
3. The Profinet to Modbus RTU gateway needs to be dragged into slot 17 to be used.
4. For detailed usage instructions, please refer to the Profinet Gateway Manual : https://www.ebyte.com/product/2341.html .

7 Precautions

(1) Do not connect the equipment while it is powered on, otherwise there is a risk of damage to the equipment.

(2) Firmware version compatibility requirements between host and expansion modules

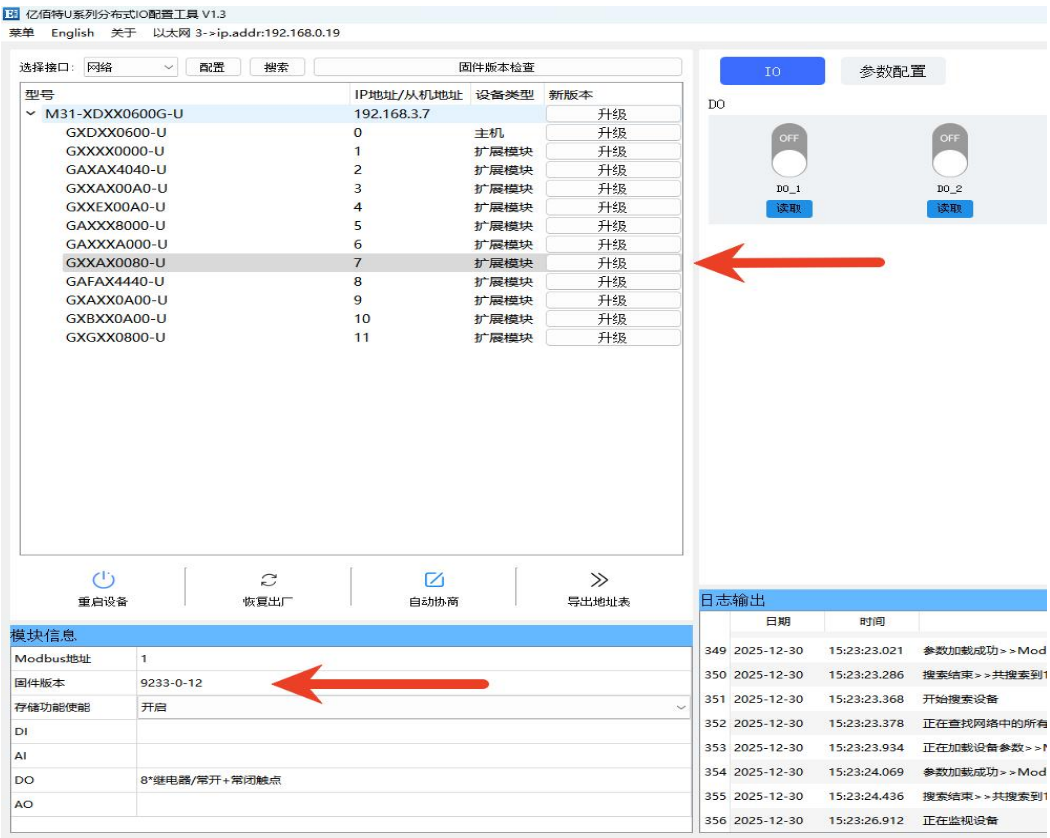
1. Compatibility principle: The firmware version number of the host device must be greater than or equal to the firmware version number of the expansion module it is paired with; otherwise, the device will not function properly.

2. Version compatibility example:

Host firmware version	Compatible expansion module firmware version
9232-0-10	9206-0-10, 9233-0-11
9232-0-11	9206-0-10, 9233-0-11, 9233-0-12
9232-0-12	9206-0-10, 9233-0-11, 9233-0-12, 9286-0-10
9232-0-13	9206-0-10, 9233-0-11, 9233-0-12, 9286-0-10, 9304-0-10

(3) Troubleshooting: If incompatible firmware versions are used in a combination, the entire device will fail to boot and operate. The host firmware must be upgraded to a version that meets compatibility requirements before normal use can proceed.

(4) Firmware version check: You can view their firmware version by selecting the host or expansion module in the host computer.



Revision History

Version	Revision Date	Revision Notes	Maintainer
1.0	2025-5-6	Initial version	LT
1.1	2025-10-11	Add new models	LT
1.2	2025-10-30	Correct port identifier	LT
1.3	2026-1-29	Add new models , modify appearance	LT

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